

Parallel Binary Refinement of Labelled Set Partitions: Exact Kernels, Spectrum, and the Last-Collision Threshold

Route-A Dynamics Working Note C301

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Abstract

We solve a synchronous fragmentation chain on the labelled set partitions of $[n]$. At every step each label receives a fresh fair bit and each current block is refined by its two nonempty bit fibres. A coupling by binary words gives the full t -step kernel from every starting partition, the complete law of the number of blocks, and the exact absorption-time distribution. The transition matrix has eigenvalues 2^{k-n} with Stirling multiplicities $S(n, k)$. A refinement flag and a genuine lowering argument give a squarefree annihilating polynomial, closing the diagonalizability gap that triangularity alone would leave. The absorption window is the birthday window: $n^2/2^t \rightarrow \lambda$ gives the limit $e^{-\lambda/2}$, with its dyadic lattice boundary kept explicit.

1 The labelled parallel-fragmentation chain

Let $n \geq 1$ and let $\mathfrak{P}_n$ be the set of labelled set partitions of $[n] = \{1, \dots, n\}$. Write $\pi \preceq \sigma$ when σ refines π . Starting from $\Pi_0 = \pi$, every label independently receives a fresh bit in $\{0, 1\}$. In each block of π , labels with equal bits stay together; empty fibres are deleted. The resulting partition is Π_1 . Updates use fresh independent bits.

This is a synchronous chain: all blocks are marked in parallel. Labels are part of the state. In particular, quotienting by block sizes produces a different chain and different spectral multiplicities. Put

$$(q)_r = q(q-1) \cdots (q-r+1), \quad (q)_0 = 1,$$

and let $S(n, k)$ denote a Stirling number of the second kind. For $\pi \preceq \sigma$, let $r_B(\pi, \sigma)$ be the number of σ -blocks contained in $B \in \pi$.

2 Complete law

Theorem 1 (all-parameter solution). *For every $n \geq 1$, $t \geq 0$, and $\pi, \sigma \in \mathfrak{P}_n$, with $q = 2^t$,*

$$K_n^t(\pi, \sigma) = \mathbf{1}_{\{\pi \preceq \sigma\}} \prod_{B \in \pi} \frac{(q)_{r_B(\pi, \sigma)}}{q^{|B|}}. \quad (1)$$

Consequently, the one-step probability equals $2^{|\pi|-n}$ precisely when σ refines π and each π -block contains at most two σ -blocks, and equals zero otherwise.

From the one-block initial partition, for every $\sigma \in \mathfrak{P}_n$ and $1 \leq k \leq n$,

$$\Pr\{\Pi_t = \sigma\} = \frac{(2^t)^{|\sigma|}}{2^{tn}}, \quad (2)$$

$$\Pr\{|\Pi_t| = k\} = S(n, k) \frac{(2^t)^k}{2^{tn}}, \quad (3)$$

$$\mathbb{E}|\Pi_t| = 2^t [1 - (1 - 2^{-t})^n]. \quad (4)$$

Let $T_n = \min\{t : \Pi_t \text{ is discrete}\}$. Then

$$\Pr\{T_n \leq t\} = \frac{(2^t)_n}{2^{tn}}, \quad (5)$$

$$\Pr\{T_n = t\} = \frac{(2^t)_n}{2^{tn}} - \mathbf{1}_{\{t \geq 1\}} \frac{(2^{t-1})_n}{2^{(t-1)n}}, \quad (6)$$

$$\mathbb{E}T_n = \sum_{t \geq 0} \left[1 - \frac{(2^t)_n}{2^{tn}} \right]. \quad (7)$$

The series converges. The characteristic polynomial, source spectral determinant, and power traces are

$$\chi_{K_n}(x) = \prod_{k=1}^n (x - 2^{k-n})^{S(n,k)}, \quad (8)$$

$$\det(I - zK_n) = \prod_{k=1}^n (1 - z2^{k-n})^{S(n,k)}, \quad (9)$$

$$\mathrm{tr}(K_n^t) = \sum_{k=1}^n S(n,k) 2^{t(k-n)}. \quad (10)$$

Moreover,

$$\prod_{k=1}^n (K_n - 2^{k-n}I) = 0, \quad (11)$$

so K_n is diagonalizable over $\mathbb{Q}$.

Finally, if $n_j \rightarrow \infty$, t_j are integers, and $n_j^2/2^{t_j} \rightarrow \lambda \in (0, \infty)$, then

$$\Pr\{T_{n_j} \leq t_j\} \longrightarrow e^{-\lambda/2}. \quad (12)$$

Thus T_n has a tight $O(1)$ window around $2 \log_2 n$.

3 Binary-word proof

After t updates, attach to label i the word formed by its successive bits. These words are independent and uniform in a set of size $q = 2^t$. Inside each starting block B , two labels are together at time t exactly when their words agree. To realize r_B prescribed terminal fibres one must injectively assign words to those fibres. There are $(q)_{r_B}$ assignments. Starting blocks use disjoint label sets, hence their counts multiply, proving (1). Taking $t = 1$ gives the stated one-step kernel.

For a one-block start, a fixed k -block partition has probability $(q)_k/q^n$. There are $S(n,k)$ such labelled partitions, yielding (2) and (3); the identity $\sum_k S(n,k)(q)_k = q^n$ also verifies normalization. Alternatively, $|\Pi_t|$ is the number of occupied boxes after n independent throws into q boxes. Summing the q occupancy indicators proves (4).

Absorption means that all n words are distinct, which proves (5); differencing successive distribution functions gives (6). The tail-sum formula gives (7). A union bound gives

$$\Pr\{T_n > t\} \leq \binom{n}{2} 2^{-t},$$

so the tail series converges. For $n = 1$, all formulas reduce to $T_1 = 0$. The contract begins at $n = 1$; the empty-set convention $n = 0$ is not used.

4 The spectral flag and its missing step

Order states by their number of blocks. Refinement makes K_n triangular. A k -block state stays unchanged exactly when every block receives a constant bit; the 2^k successful assignments among 2^n have probability $\lambda_k = 2^{k-n}$. Hence the diagonal rank- k block is $\lambda_k I_{S(n,k)}$, proving (8); (9) and (10) follow once diagonalizability is closed.

Triangularity alone is not that closure. Let V_k be the span of basis vectors indexed by partitions of rank at most k , with the conventional column action of the displayed transition matrix. The spaces

$0 \subset V_1 \subset \cdots \subset V_n$ are invariant, and $(K_n - \lambda_k I)V_k \subseteq V_{k-1}$. Since all polynomial factors in K_n commute, applying the factors in descending rank lowers this flag one step at a time. Their product kills V_n , proving (11). The roots $\lambda_1, \dots, \lambda_n$ are distinct, so the minimal polynomial is squarefree and K_n is diagonalizable over $\mathbb{Q}$. This also justifies the trace formula without any hidden Jordan-block assumption.

5 Birthday threshold and lattice boundary

Write $q_j = 2^{t_j}$. Under the hypotheses of (12), $n_j/q_j \rightarrow 0$ and

$$\begin{aligned} \log \frac{(q_j)_{n_j}}{q_j^{n_j}} &= \sum_{r=0}^{n_j-1} \log(1 - r/q_j) \\ &= -\frac{n_j(n_j-1)}{2q_j} + O\left(\frac{n_j^3}{q_j^2}\right) \rightarrow -\lambda/2. \end{aligned}$$

Exponentiation proves (12). The same product, together with the union bound above, yields tightness around $2 \log_2 n$.

There is an essential lattice qualification. Because t is integer and $q = 2^t$, the quantity $n^2/2^{\lfloor 2 \log_2 n + c \rfloor}$ need not converge without controlling its dyadic phase. We therefore assert the limit only along sequences for which that ratio converges; no phase-free continuous law for $T_n - 2 \log_2 n$ is claimed.

References

- [1] P. Diaconis, C. Y. A. Pang, and A. Ram, “Hopf algebras and Markov chains: two examples and a theory,” *Journal of Algebraic Combinatorics* (2014), doi:10.1007/s10801-013-0456-7; arXiv:1206.3620.